Quantitative text analysis – Essex Summer School

Topic models, LDA, STM, and seeded topics

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Today's class

- Introduction to unsupervised and semi-supervised topic models
  - Latent Dirichlet Allocation (LDA)
  - Structural Topic Models (STM)
  - Seeded LDA and LLM-assisted topic labelling
- Lab session
  - Topic model exploration, labelling, and visualisation with House of Commons speeches

House of Commons speeches

- We continue with the same House of Commons leader speech sample from Lecture 6:
  - Labour and Conservative leaders from 1979–2025
  - up to 200 speeches per leader in each legislative period
  - individual speeches retain party, leader, date, agenda, and election period metadata
- The source corpus is the same, but this time we'll keep individual speeches as documents
- Core question: do discovered topics look like policy agendas, procedural language, time periods, or leader/party style?

Questions addressed by topic models

Our sample lets us ask several kinds of questions that we could answer using topic models.

Question	Evidence from topic model output
Which themes appear in each speech?	document-topic proportions
Which words define each theme?	topic-word probabilities
How do topics vary by leader, party or election period?	metadata-based comparisons

Why topic models?

- Topic modelling helps us discover latent themes in a text corpus *without pre-specifying categories*
- Useful for:
  - Summarising large collections of text
  - Exploring new corpora
  - Informing downstream tasks like classification or measurement
- But keep in mind your research goal: discovery vs. measurement (Grimmer, Roberts, & Stewart, 2022)
  - Discovery: generating new hypotheses or understanding the corpus
  - Measurement: using topics as variables – requires careful validation and alignment with theoretical constructs

What topic model output looks like

From the HoC leader speech sample, a six-topic model produces outputs like this:

Tentative Label	High-Probability Terms
Prime ministerial statements and inquiries	prime, said, government, inquiry, queen, lord
Health and public services	people, health, service, government, nhs, care
Europe and the EU	european, union, europe, eu, deal, countries
Economy and budget	government, tax, chancellor, budget, year, per_cent
Security and foreign affairs	security, support, british, world, military, forces
Constitutional/procedural politics	northern, ireland, debate, scotland, leader

Important: the labels are our interpretation. The model gives us terms, topic proportions, and high-topic speeches.

Inspecting a Topic

For the fitted Security and foreign affairs topic, the speeches with the highest topic proportion include:

Speech	Party	Agenda	Proportion
Tony Blair, 2006	Labour	PM engagements	0.95
Iain Duncan Smith, 2003	Conservative	PM engagements	0.93
Harriet Harman, 2010	Labour	PM engagements	0.90
Rishi Sunak, 2023	Conservative	Israel and Gaza	0.87
Keir Starmer, 2024	Labour	Iran-Israel update	0.86

This is the basic interpretive workflow: top words plus high-topic speeches.

Topic models

Research goal: find a number of k topics, consisting of specific words and/or documents, that minimise the mistakes we would make if we try to reconstruct the corpus from the topics (Van Atteveldt *et al.* 2022, p. 203)

- There are many different topic models, which share the following characteristics (Grimmer & Stewart, 2013):
  - Each topic is a probability distribution over features
  - The model assumes a generative process for observed text
- Some topic models are "mixture models" (a document can consist of multiple topics), others allocate each document to one topic only (single membership model)

Why focus on LDA?

Why focus on Latent Dirichlet Allocation (LDA; Blei, Ng & Jordan, 2003)?

- Widely used, flexible, and relatively intuitive output:
    - Topic distribution per document – how much of each topic appears in each document
    - Word distribution per topic – what words define each topic
    - E.g., “Document A is 80% topic 1, 15% topic 2; Document B is 60% topic 3...”
- Serves as a foundation for many extensions:
    - Correlated Topic Models (CTM)
    - Dynamic Topic Models (DTM)

Assumptions behind LDA

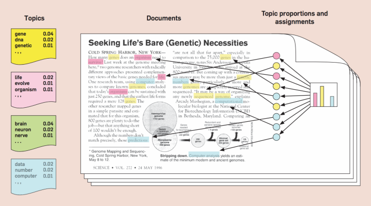
- Documents are modeled as a mixture of topics, and topics as a mixture of words
  - Each word in a document is drawn from one of the topics assigned to that document
  - This is a mixed membership model
- Number of topics k is fixed in advance (needs to be chosen by the researcher)
- LDA is a probabilistic model:
  - Running the same model multiple times may yield different results due to random initialisation
  - This is known as the multimodality problem (Roberts *et al.*, 2016)
  - Solutions include multiple runs, stability checks, and model validation

The generative process of text

For each **document** in the corpus, **words** are generated in a two-stage process:

1. Randomly choose a distribution over topics
2. For each **word** in a **document**
   - Randomly choose a topic from the distribution over topics in step # 1
   - Randomly choose a word from the corresponding distribution over the vocabulary in that topic

Figure 1. The intuitions behind latent Dirichlet allocation. We assume that some number of "topics," which are distributions over words, exist for the whole collection (far left). Each document is assumed to be generated as follows. First choose a distribution over the topics (the histogram at right); then, for each word, choose a topic assignment (the colored coins) and choose the word from the corresponding topic. The topics and topic assignments in this figure are illustrative—they are not fit from real data. See Figure 2 for topics fit from data.



Inference: from words to topics

- The only thing we observe is words
  - Our document-feature matrix
- Given our generative process, we infer the topic structure that is most likely to have generated the observed words
  - Move in the opposite direction of the generative process
- To start off this inferential process we need to make some assumptions:
  - A prior distribution of topics per document θ
  - A prior distribution for words across topics β

How LDA Learns Topics

LDA starts with these generative assumptions, observes the corpus, and updates its estimate of the topic structure.

Object	What it describes
θ	topic mixture in each speech
β	word probabilities within each topic
α	how mixed or focused speeches are allowed to be
η	how broad or focused topic vocabularies are allowed to be

Low α means speeches are expected to concentrate on fewer topics. Low η means topics are expected to concentrate on fewer words.

In practice: LDA repeatedly asks whether a word fits better with a different topic because that topic is common in the speech and because the word is common in that topic.

Document-topic distribution θ

Rows are speeches; columns are topics; cells are topic proportions.

Speech	PM/inquiry	Health	Europe	Budget	Security	Procedure
Thatcher, 1979	0.01	0.02	0.00	0.15	0.00	0.82
Blair, 2006	0.02	0.01	0.00	0.01	0.95	0.01
Brown, 2010	0.02	0.03	0.00	0.87	0.04	0.03
Starmer, 2024	0.04	0.02	0.01	0.03	0.86	0.04

Each row sums to approximately 1. A speech can be mostly one topic or a mixture of several topics.

Word-topic distribution β

Rows are words; columns are topics; cells are word probabilities within topics.

Word	PM/inquiry	Health	Europe	Budget	Security	Procedure
prime	high	low	low	low	low	low
nhs	low	high	low	low	low	low
european	low	low	high	low	low	low
tax	low	low	low	high	low	low
military	low	low	low	low	high	low
ireland	low	low	low	low	low	high

These word probabilities are why top-word lists are the first clue for topic interpretation.

Inference, Intuitively

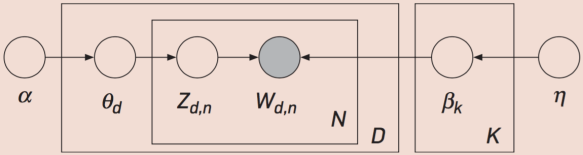
We do not observe topics directly. We observe words in speeches and infer which topic assignments would make those words plausible.

1. Start with provisional topic assignments for words.
2. Reassign words using two pieces of information:
   - is this topic common in this speech?
   - is this word common in this topic?
3. Repeat until the topic-word and document-topic patterns stabilize.

Different software uses different approximate inference strategies, such as Gibbs sampling or variational inference.

Graphical model for LDA (Blei, 2012)

Figure 4. The graphical model for latent Dirichlet allocation. Each node is a random variable and is labeled according to its role in the generative process (see Figure 1). The hidden nodes—the topic proportions, assignments, and topics—are unshaded. The observed nodes—the words of the documents—are shaded. The rectangles are "plate" notation, which denotes replication. The *N* plate denotes the collection words within documents; the *D* plate denotes the collection of documents within the collection.



Determining k

- Number of topics k is chosen by the researcher
- There is rarely a single “correct” value
- Useful diagnostics:
  - Interpretability: can we give topics defensible labels?
  - Coherence: do top words belong together?
  - Exclusivity: are topics distinct from one another?
  - Stability: do similar topics appear across reasonable runs?
  - Prediction: does perplexity improve on held-out text?

For social science: prefer a model that is interpretable, stable, and useful for the research question over a model that only optimizes prediction.

Validating topic models in context of measurement

- Semantic validity: extent to which topics are coherent
  - Absence of random, unrelated words
  - Topics that are specific enough and not overly general
  - Can be evaluated using coders – check out the **oolong** library in R (Chan & Sältzer, 2020)
- Predictive validity: how well variation in topic usage corresponds with predicted events
  - E.g., a terrorism topic in media reports should peak after a terrorist incident
- Convergent validity: extent to which model output can be validated with other approaches
- Birkenmaier et al. (2024) refer to this as internal and external validation

LDA in the HoC lab

- We start with a plain unsupervised LDA model
- We choose $k = 6$ topics, inspect terms, inspect θ, and read high-topic speeches
- This gives us the baseline workflow before adding metadata in STM or seed words in seeded LDA

```r
lda_6 <- textmodel_lda(
  topic_dfm,
  k = 6,
  alpha = 0.5,
  max_iter = 500
)

terms(lda_6, 10)
head(lda_6$theta)
```

Structural topic models

We often have metadata in our corpus:

- For a leader speech corpus: *speaker, party, year, debate agenda, etc.*

Structural topic models (Roberts *et al.*, 2014) allow a topic model to use that data to infer topical content and topical prevalence

- Accompanying website structuraltopicmodel.com provides extensive resources and vignettes

Original application was on open-ended survey data in the US political context. Do demographics and partisan preferences of respondents affect their responses?

STM generative process

- Topics are initialised deterministically (if you run the same STM on the same data twice you get the same outcome)
- Topic proportions θ drawn from multinomial logistic normal distribution with covariates
  - Topical prevalence per document can correlate with covariates
- Topic words β also drawn from multinomial logistic normal distribution with covariates
  - Topical content can correlate with covariates

STM in the HoC lab

- We estimate six topics and model topic prevalence as a function of party and election period
- The output is still interpreted through top terms, FREX terms, and high-topic documents
- The metadata model helps us ask: which topics are more common for Labour or Conservative leaders, and which topics vary over time?
- This matters because a topic can be substantively coherent but still be driven by party, period, agenda, or speaker role

```r
stm_6 <- stm(
  topic_dfm,
  data = docvars(topic_dfm),
  K = 6,
  prevalence = ~ party_family + election_no,
  seed = 20260714,
  max.em.its = 25,
  init.type = "Spectral",
  verbose = FALSE
)
```

Semi-supervised topic models

Some topic models are semi-supervised: they rely on an *ex ante* mapping of words to topics, combining inductive and deductive aspects.

- **keyATM** (Eshima *et al.* 2023) and **seeded LDA** (Watanabe & Baturo, 2023)

Workflow: choose keywords for theoretically grounded topics and use them to inform some or all topics.

- Quality of the topics depends in large part on the quality of the seed words
- Difference from a dictionary:
  - a dictionary directly counts seed terms
  - seeded LDA uses seeds to guide topic discovery and can learn additional related words from the corpus
- LLMs can suggest seeds, but check suggestions against corpus language and theory

Seeded LDA in the HoC lab

- We seed broad policy topics that should plausibly occur in parliamentary speeches
- Residual topics can capture language that does not fit the seeded categories

```
hoc_topic_dictionary <- dictionary(
  list(
    economy = c("econom*", "tax*", "budget*", "inflation", "growth", "jobs"),
    health = c("health*", "nhs", "hospital*", "patient*", "care"),
    security = c("security", "defence", "police", "crime", "border*"),
    environment = c("climate", "environment*", "carbon", "energy", "green")
  )
)

seeded_topics <- textmodel_seededlda(
  topic_dfm,
  hoc_topic_dictionary
)
```

Other types of BoW topic models

- Many other topic models vary the assumptions:
  - **Correlated topic models** (Blei & Lafferty, 2005): topics can be correlated
  - **Dynamic topic models** (Blei & Lafferty, 2006): topical content and prevalence can vary over time
  - **Hierarchical topic models** (e.g., Grimmer, 2010): researcher does not fix k in advance
- Accessibility in R varies, so workflow choices are partly practical

Embedding-based topic models

- Main idea: represent each document as a dense vector, then look for groups of semantically similar documents
- Common workflow, e.g. BERTopic:
    1. embed documents with a language model
    2. cluster nearby documents
    3. extract keywords to describe each cluster
- Compared with LDA, this is less dependent on exact word overlap and can group texts that use different words for similar ideas
- But keyword summaries often still use word-frequency-style evidence
- Validate labels, inspect documents, and check whether clusters reflect concepts rather than style, period, speaker, or genre

LLMs and topic labelling

- Topic labels are interpretations, not model outputs
- LLMs can help by:
  - proposing short labels from top words and example documents
  - comparing alternative labels
  - flagging topics that mix several policy areas
- But the final label should be checked using:
  - high-probability speeches
  - party, leader, agenda, and election-period patterns
  - sensitivity to preprocessing, k, and seed words